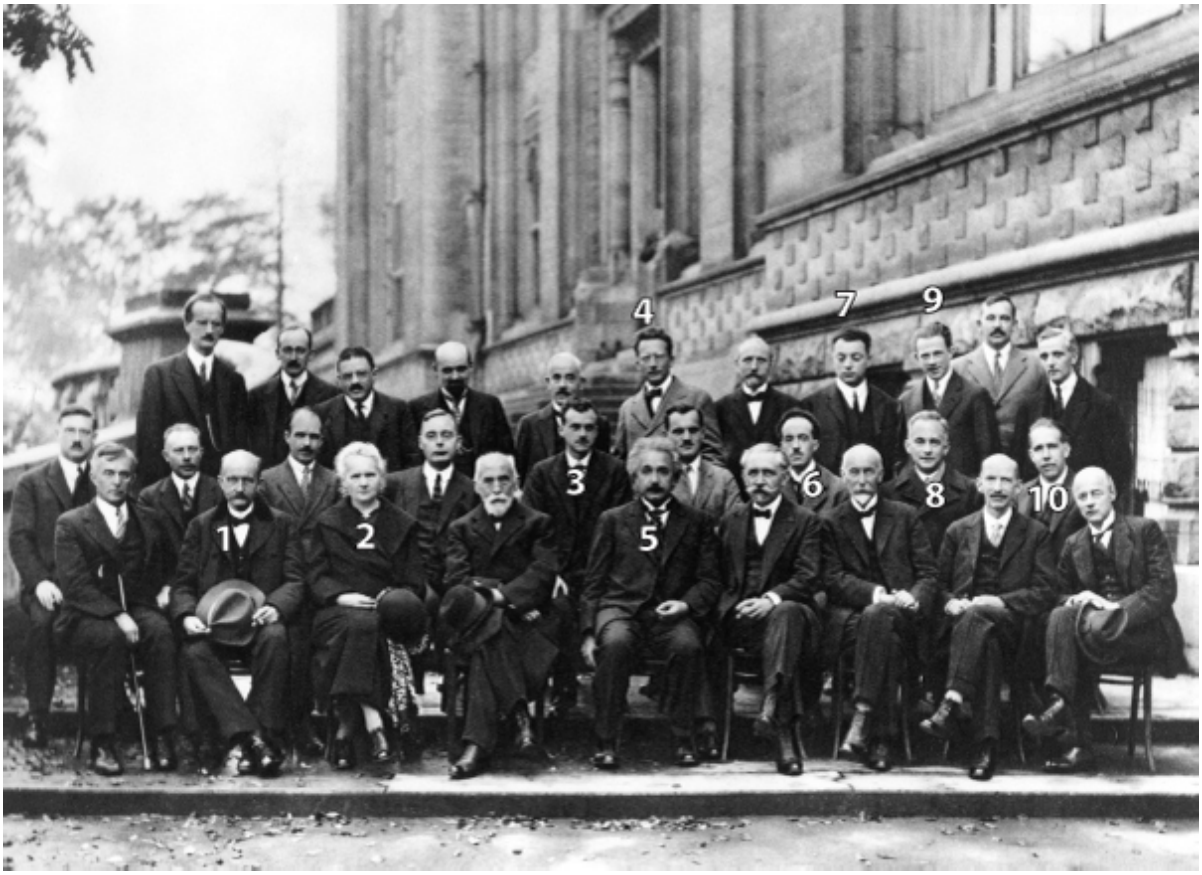


Participants in the 1927 Solvay Conference. Among the more well-known were: 1. Max Planck, 2. Marie Curie, 3. Paul Dirac, 4. Erwin Schrödinger, 5. Albert Einstein, 6. Louis de Broglie, 7. Wolfgang Pauli, 8. Max Born, 9. Werner Heisenberg, and 10. Niels Bohr. (Courtesy of Wikipedia)

Einstein and his compatriots had reason to be cautiously optimistic that such a new-and-improved theory was out there to be found. Just a few decades before, in the later years of the nineteenth century, physicists had developed the theory of statistical mechanics, which described the motion of large numbers of atoms and molecules. A key step in that development—which all took place under the rubric of classical mechanics, before quantum theory came on the scene—was the idea that we can talk profitably about the behavior of a large collection of particles even if we don't know precisely the position and velocity of each one of them. All we need to know is a *probability distribution* describing the likelihood that the particles might be behaving in various ways.

In statistical mechanics, in other words, we think that there actually is some particular classical state of all the particles, but we don't know it, all we have is a distribution of probabilities. Happily, such a distribution is all